Paper: organic chemistry (2020+2019)
Semester: 6th (CHEM-317)

GA

Question Number: 2(I) (2020)
3(i) (2019)

Swern oxidation:

"Swern oxidation refers to the organic chemical reaction whereby a primary or secondary alcohol (-OH) is oxidized to an aldehyde (-CH=O) or ketone (>C=O) using oxalyl chloride, dimethyl sulfoxide (DMSO) and an organic base, such as triethylamine."

→ It is named after Daniel Swern, an American chemist who discovered the Swern oxidation.

→ It is one of the many oxidation reactions commonly referred to as 'activated DMSO oxidations'.

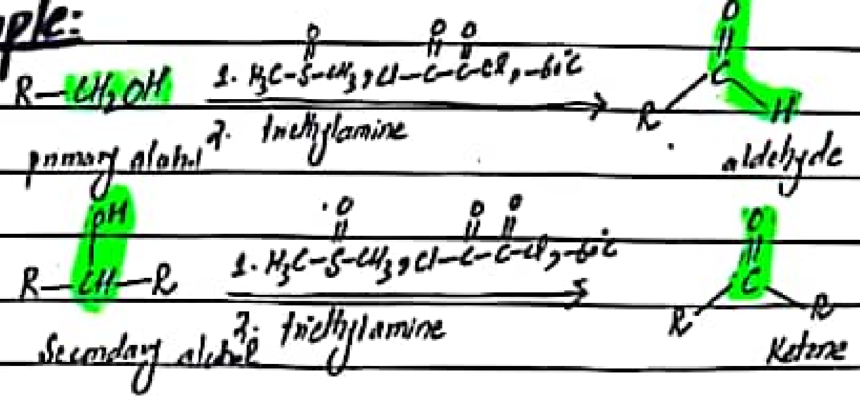
By products:

The by-products are dimethyl sulfide (Me_2S, also abbreviated as DMS), carbon monoxide (CO), carbon dioxide (CO_2) and — when triethylamine is used as base — triethylammonium chloride (Et_3NHCl).

Disadvantage:

The major disadvantage of this reaction is the formation of dimethyl sulphide as a side product, which has an unpleasant smell which can become highly disagreeable even in low concentrations.

Example:



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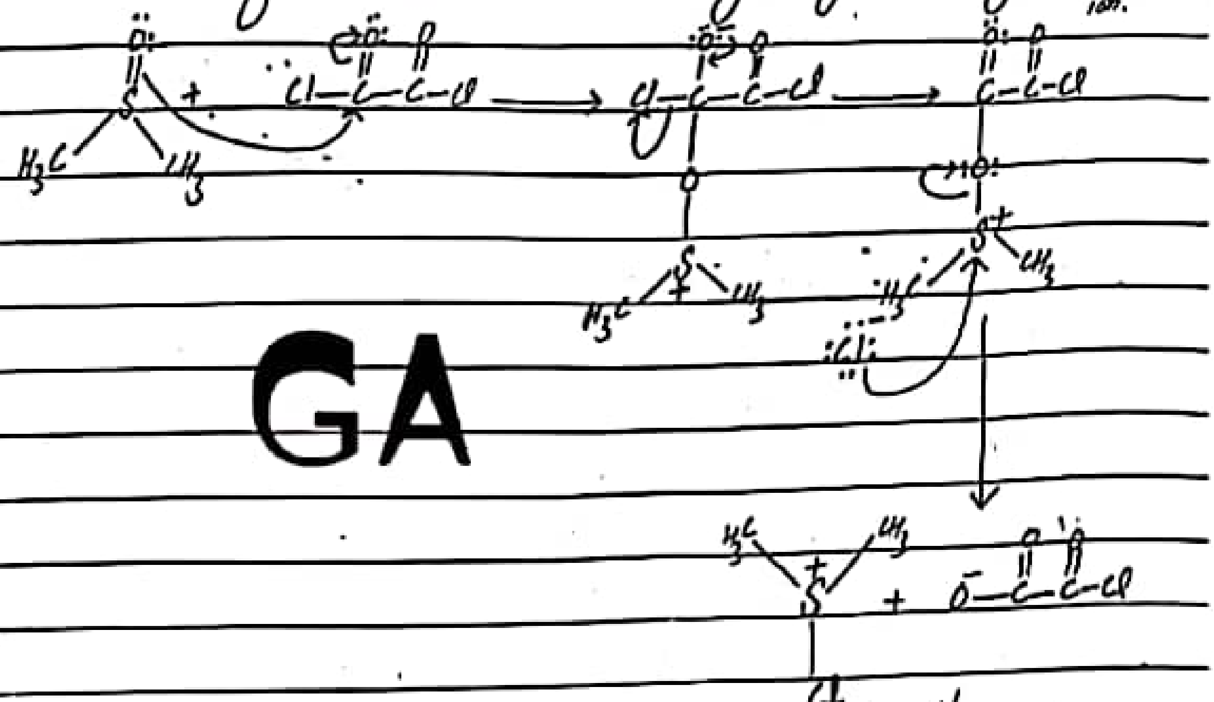
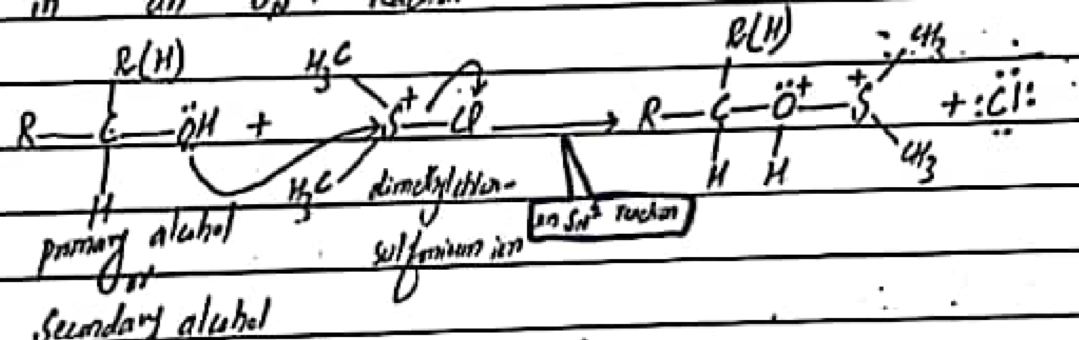
→ First of all dimethylsulfoxide [$(CH_3)_2SO$] and oxalyl chloride [$(COCl)_2$] react to form the actual oxidizing agent, dimethylchlorosulfonium ion.

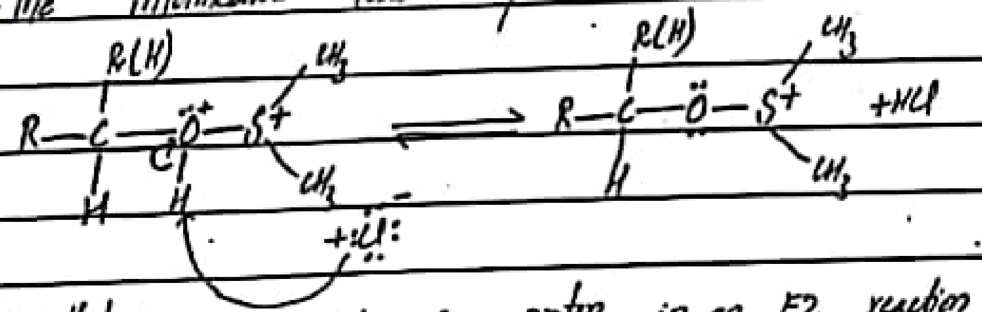


dimethylchlorosulfonium ion

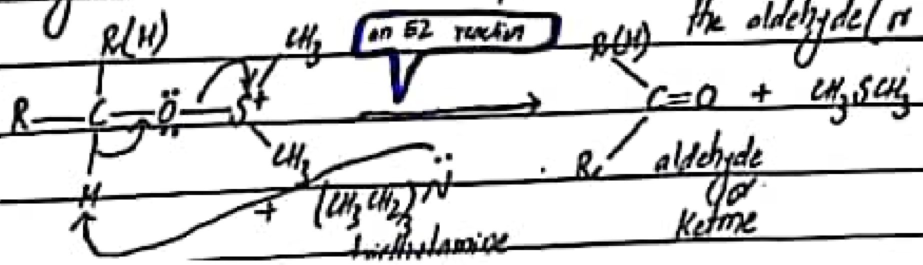
→ The alcohol displaces the chloride ion of dimethylsulfonium ion in an S_N2 reaction.



→ The intermediate loses a proton.



→ Triethylamine removes a proton in an E2 reaction to form the aldehyde (or Ketone).



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Paper: organic chemistry (2020+2019)
Semester: 6th (CHEM-317)

GA

Question No. 2(ii)(2020)
2(ii)(2019)

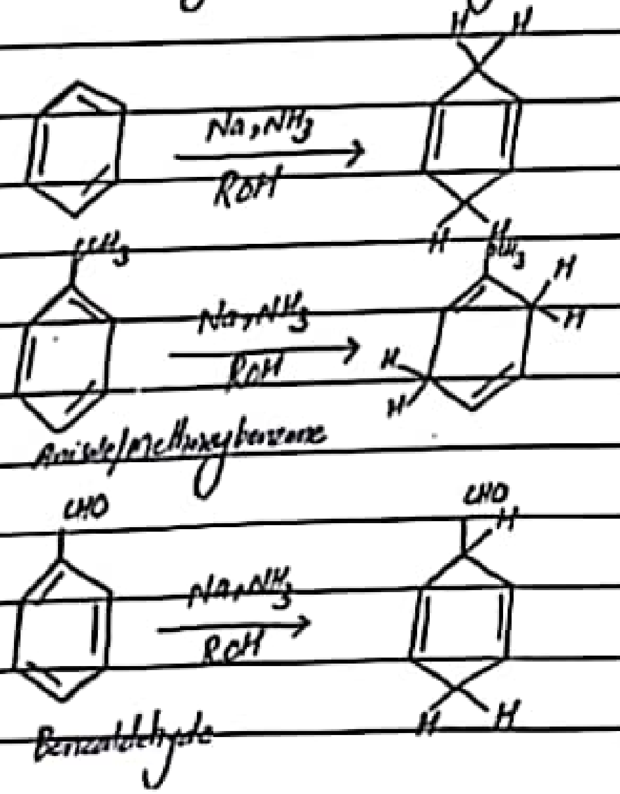
Birch reduction:

The Birch reduction is an organic reaction that is used to convert arenes to 1,4-cyclohexadienes.

→ The reaction is named after the Australian chemist Arthur Birch and involves the organic reduction of aromatic rings in an amine solvent (traditionally liquid ammonia) with an alkali metal (traditionally sodium) and a proton source (traditionally an alcohol).

→ Unlike catalytic hydrogenation, Birch reduction does not reduce the aromatic ring all the way to cyclohexene.

Examples:



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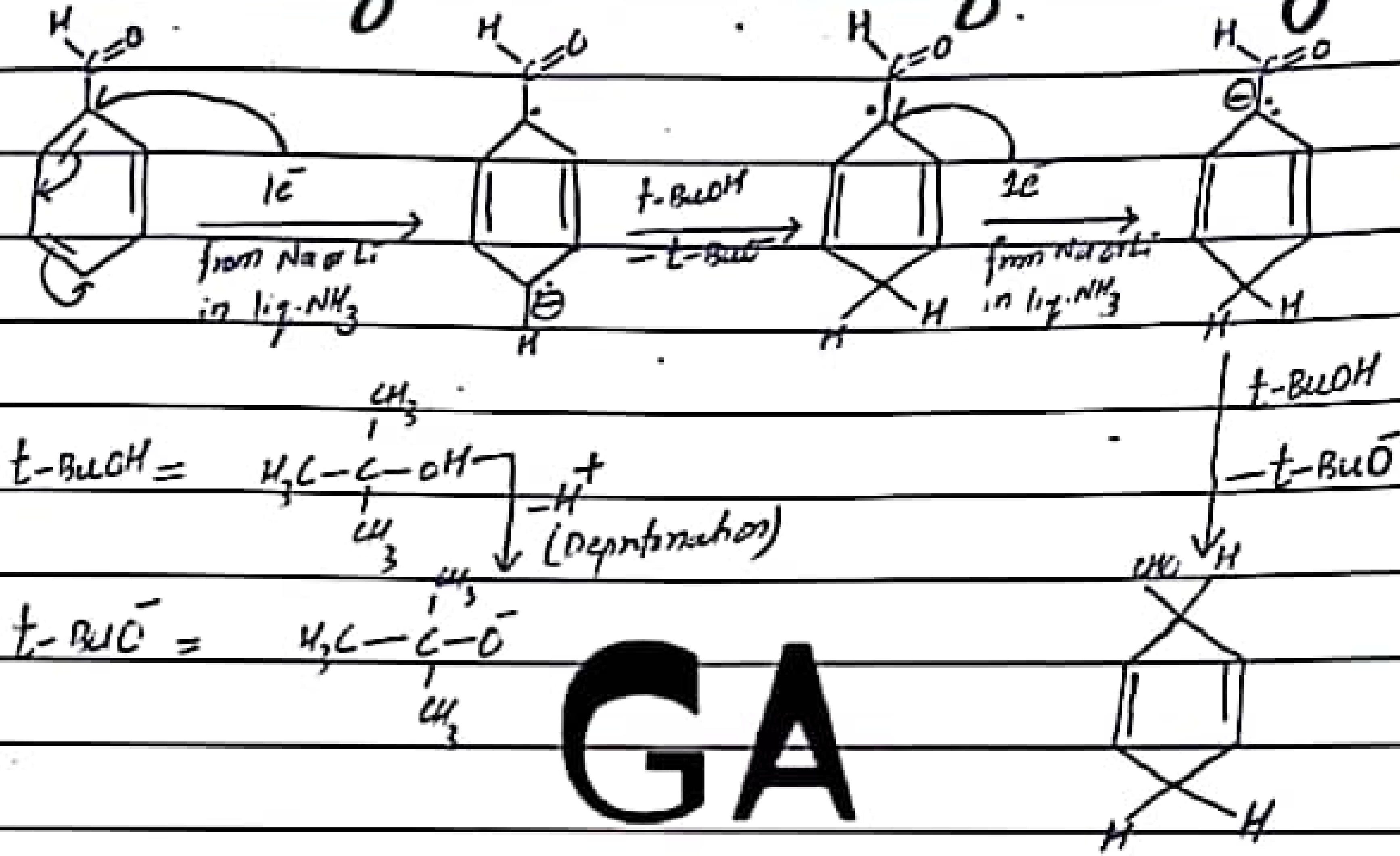
Mechanism for Birch reduction of Benzaldehyde



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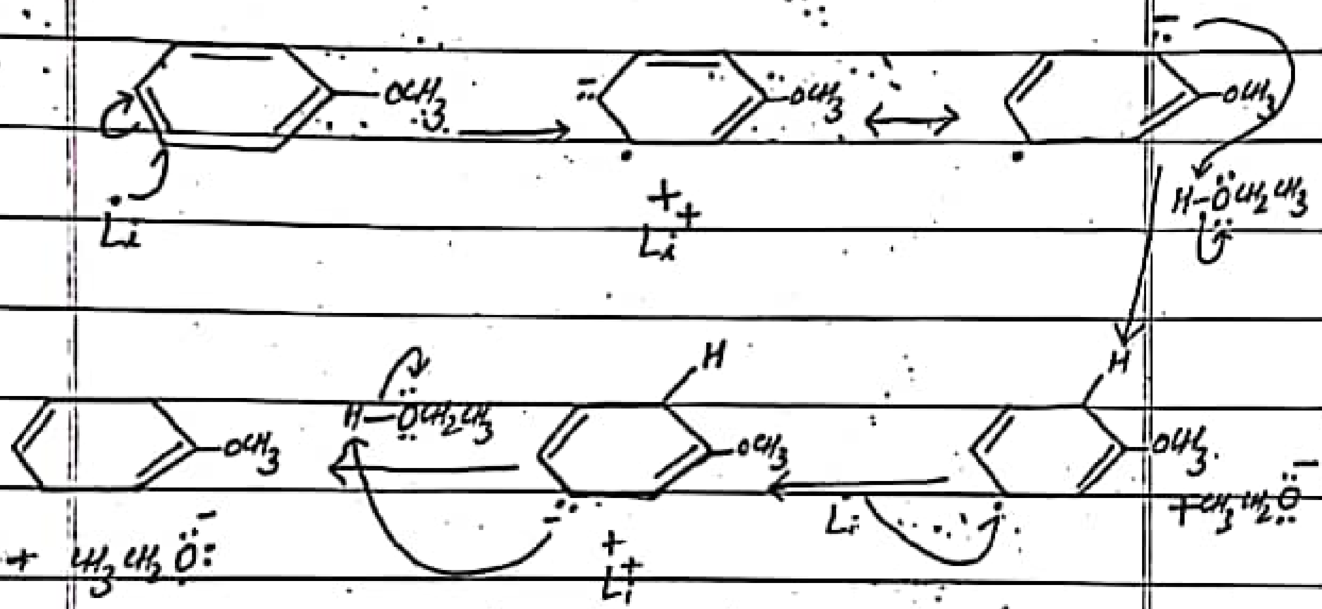
Paper : Organic Chemistry (Fall 2022)

Semester : 6th (CHEM-[redacted])

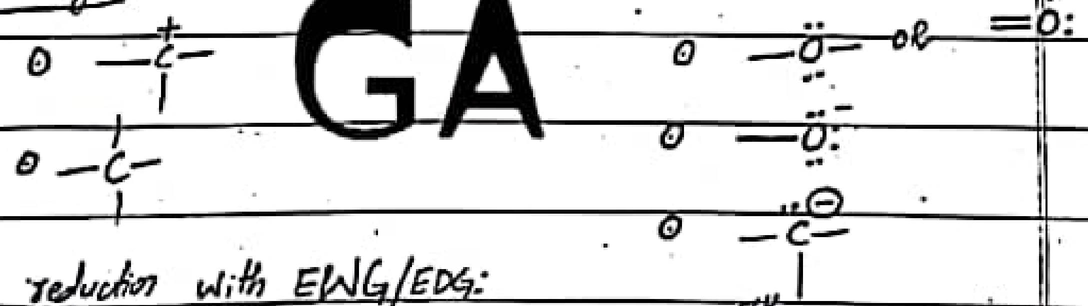
Q.NO.3

(ii)

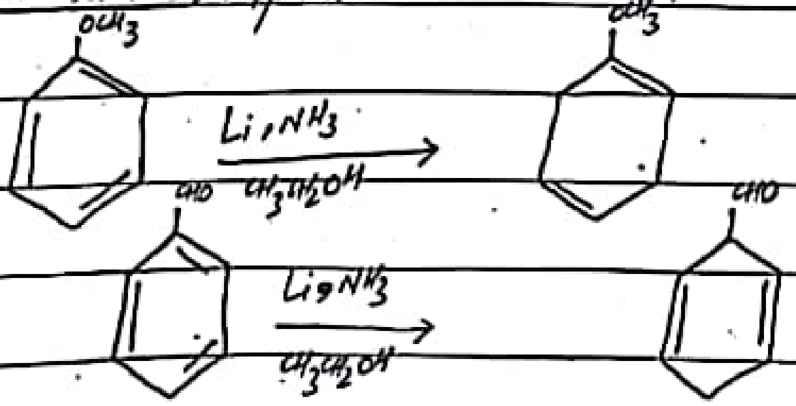


Arrow notation: 

Formal charges:



Birch reduction with EWG/EDG:



$-OCH_3$ = EDG

$-CHO$ = EWG

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Synthetic applications of free radicals

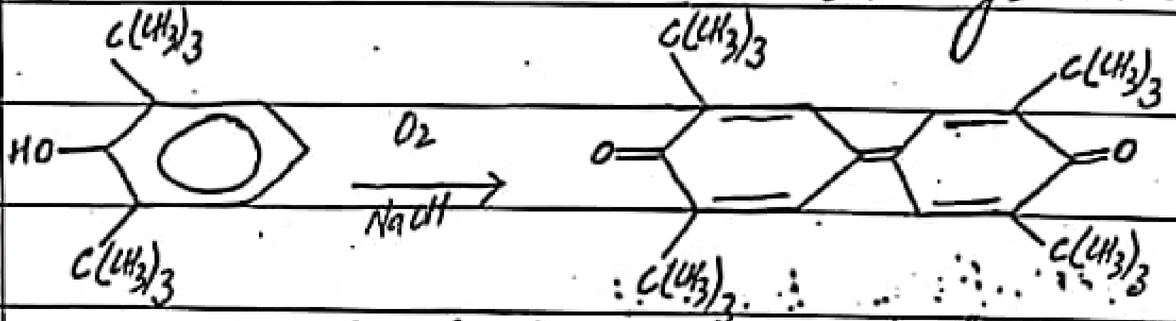
(i) Free radical polymerization is important reaction fo synthesis of many useful polymers like polystyrene, polyacetonitrile, polyvinyl chloride, polyethylene, polypropy

(ii) Many organic compounds can be synthesized by radical mechanism. For example

(a) synthesis of halo compounds

(b) 'Kolbe' electrolysis for the synthesis of carboxylic

(iii) Oxidative coupling reactions of phenols are used for synthesis some dyes and drug



2,6-Di-tert-butylphenol a biphenyl quinone (red b

Paper: Organic Chemistry (2020)
Semester: 6th (CHEM-317)

GA

Question NO: 2

(IV)

Types of vibration in IR spectroscopy:

Stretching vibrations:

A stretching vibration is movement of atoms along the bond axis such that interatomic distance is increasing or decreasing.

→ According to the direction of stretching, it can be further divided into two parts.

a) Symmetrical stretching:

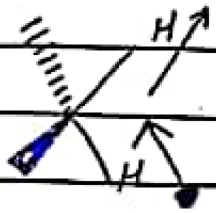
In diatomic molecules both atoms of the bond are coming near to central point (gravity centre) at the same time and go far from the centre point at the same time is known as "symmetric stretching" vibration.

→ These vibrations are generally IR inactive because it does not bring change in dipole moment.

b) Asymmetrical stretching

In diatomic molecules out of the two atoms, when one atom comes near to central point while other goes away from the central point is known as "asymmetrical stretching" vibration.

→ These vibrations required higher energy as compare to bending vibrations, and absorption occurs at a higher frequency (wave number).



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A bending vibration is correlated to change in bond angles between bonds with a common atom or the movement of a group of atoms with respect to one another.

→ These are further divided into two groups.

i) In-plane bending
ii) out-of-plane bending

i) In-plane bending:

During bending vibration, if the position of atoms change in the same plane, then it is known as in-plane bending vibrations.

→ On the basis of the motion of atoms it can be further classified into two groups.

a) Scissoring: If atoms of the molecule come near to each other or far from each other without a change in bond length as well as plane, then it is known as scissoring vibration.

b) Rocking: In this type, the movement of the atoms takes place in the same direction.

ii) Out-of-plane bending:

During bending vibrations, if the position of atoms change in the different planes, then it is known as out-of-plane bending vibrations.

a) Wagging: In this type, two atoms move up and down the plane with respect to the central atom.

b) Twisting: In this type, one of the atoms moves up the plane while the other moves down the plane with respect to the central atoms.

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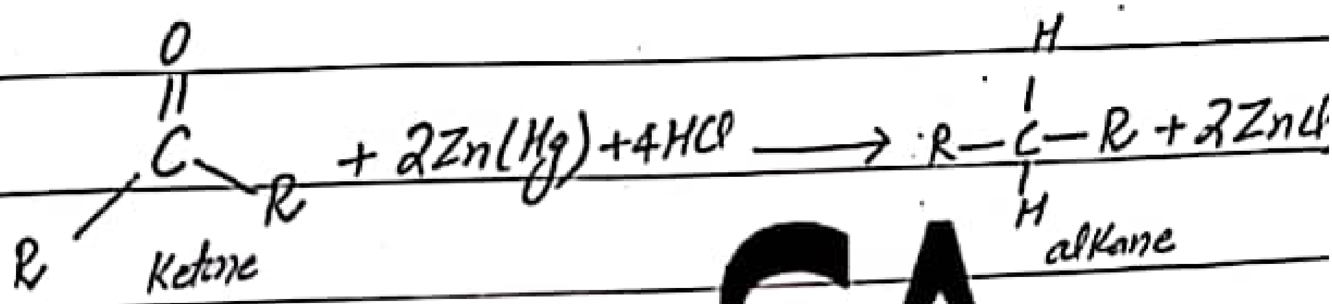
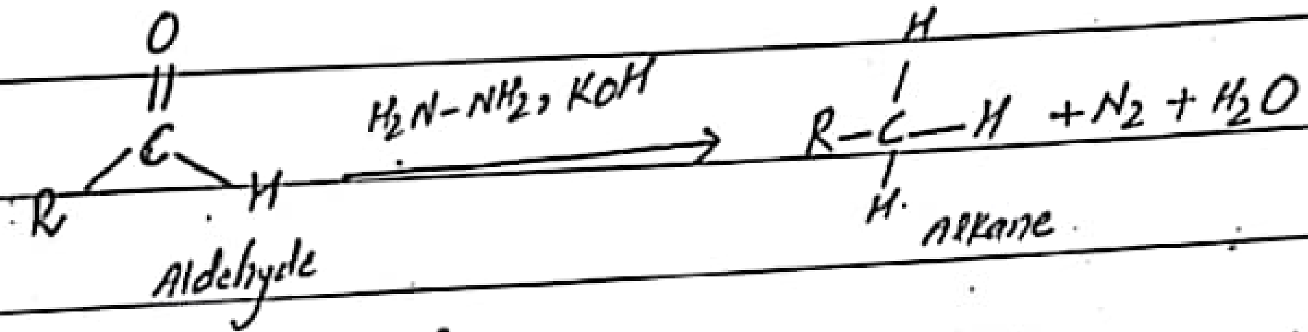
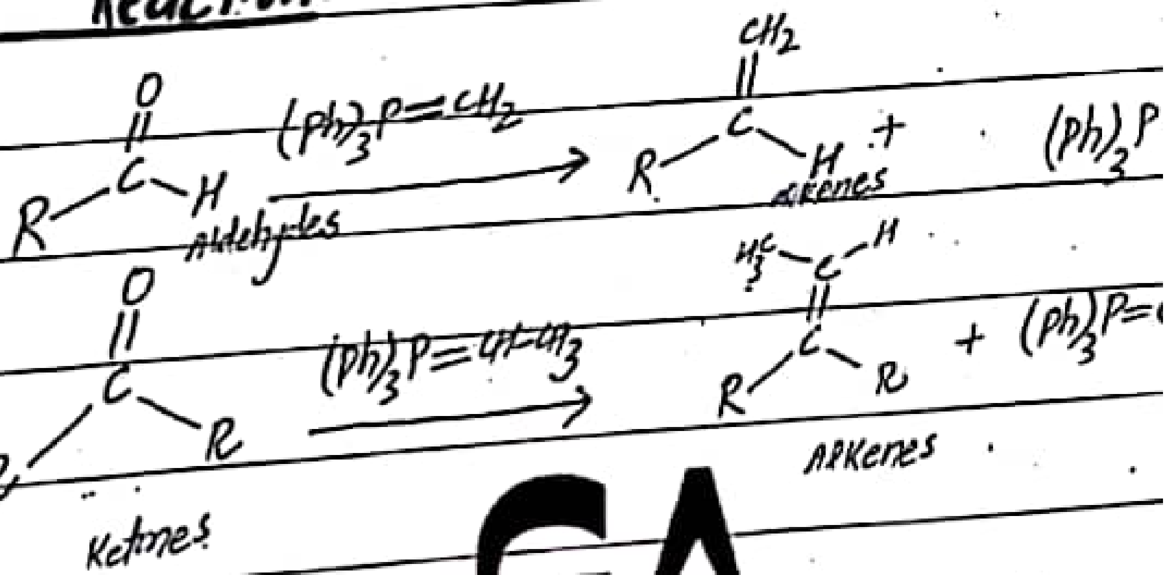
Clemmensen Reduction:



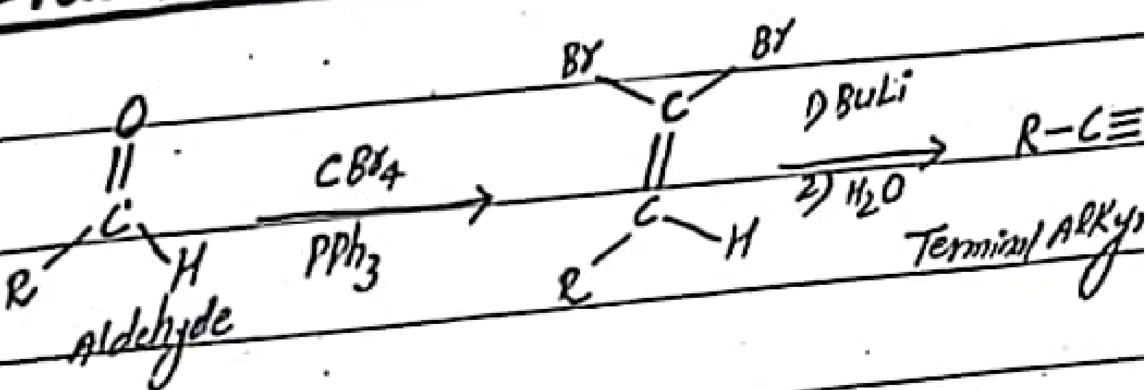
Wolff-Kishner reduction:



Witting Reaction:



Corey-Fuchs Reaction:



Paper: Organic Chemistry (2020)
Semester: 6th (CHEM-317)
Question No: 3(I)

Ozonolysis of Alkenes:

"Ozonolysis is a method of oxidatively cleaving alkenes using ozone, a reactive allotrope of oxygen."

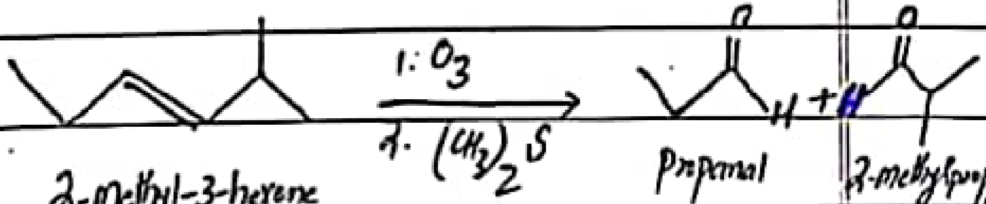
→ The process allows for carbon-carbon double bonds to be replaced by double bonds with oxygen.

→ This reaction is often used to identify the structure of unknown alkenes by breaking them down into smaller, more easily identifiable pieces.

Examples:

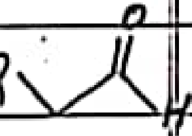
$$CH_3-\underset{|}{\overset{CH_3}{C}}=CH-CH_2-CH_3 \xrightarrow[2.\ (CH_3)_2S]{1.\ O_3} CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + H-\overset{O}{\overset{\|}{C}}-CH_2-CH_3$$

2-methyl-2-pentene → propanone + propanal



2-methyl-3-hexene → Propanal + 2-methylpropanal (isobutyraldehyde)

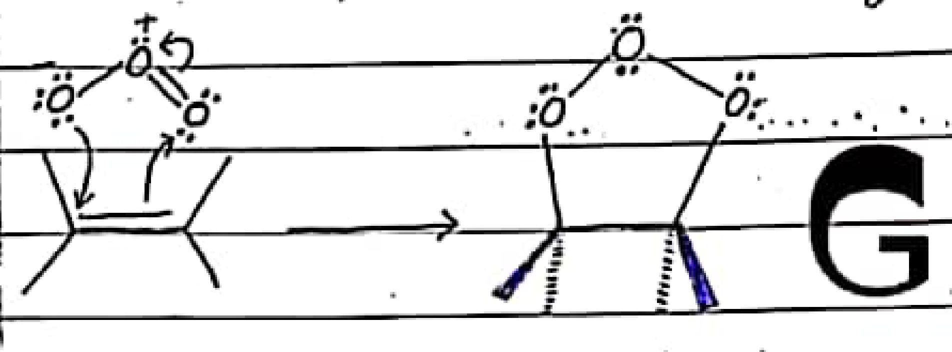
$$CH_3-CH_2-CH=CH-CH_2-CH_3 \xrightarrow[2.\ (CH_3)_2S]{1.\ O_3} 2$$



3-pentene → propanal (two moles)

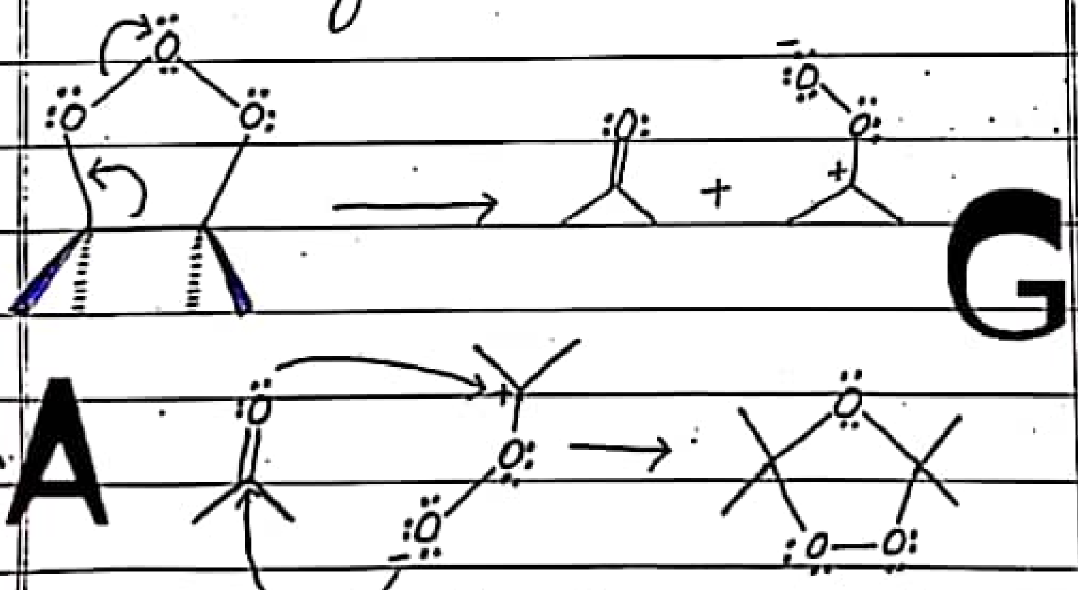
Mechanism:

Step 1: Ozone reacts with the alkene with both electrophilic and nucleophilic character in a single, simultaneous step.

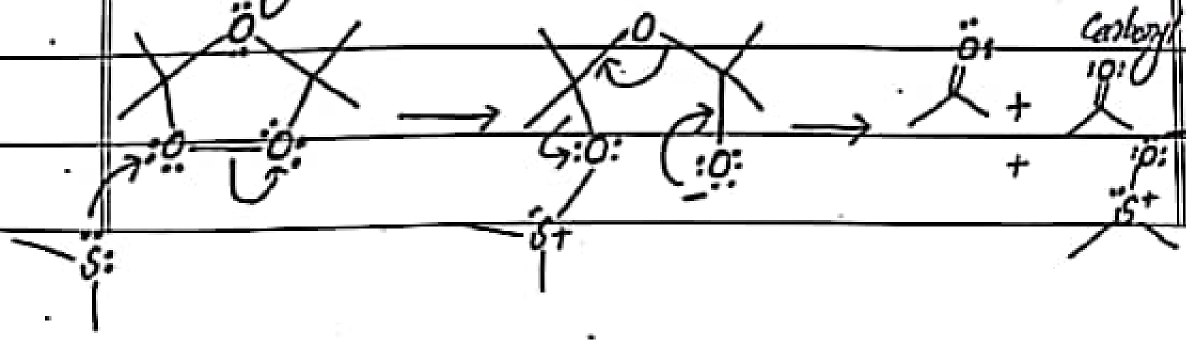


A molozonide

Step 2: Break a bond to give stable molecules or ions. Relocating valence electrons in the molozonide results in cleavage of one carbon-carbon and one oxygen—oxygen bond. The resulting fragments then recombine to form an ozonide



Step 3: Reduction of the ozonide and cleavage results in the final carbonyl fragments.



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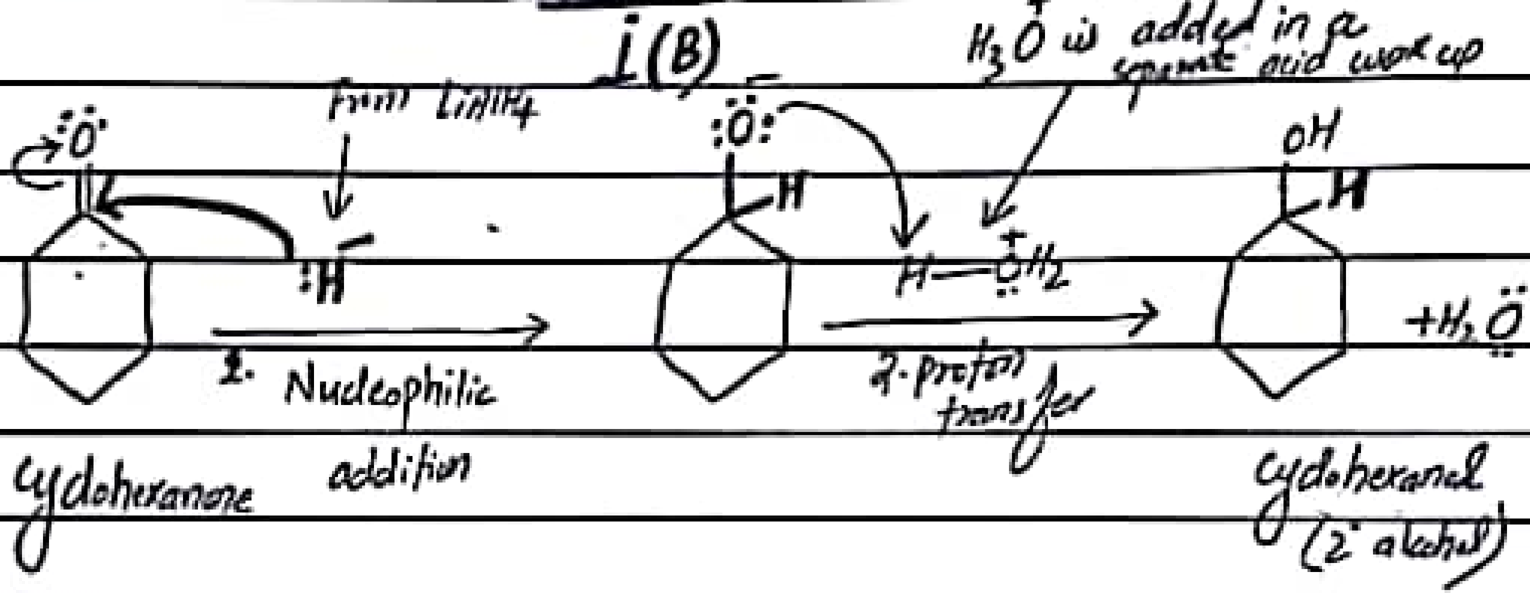
Paper : organic chemistry (2020)
Semester : 6th (CHEM-317)

GA

Q.NO.4

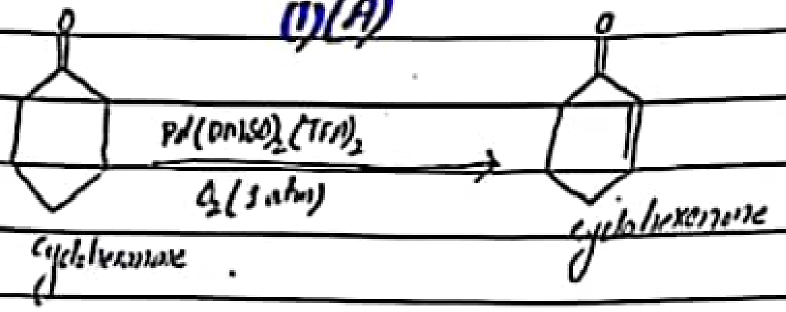
i (B)



Paper: organic chemistry (2020)
Semester: 6th. (CHEM-317)

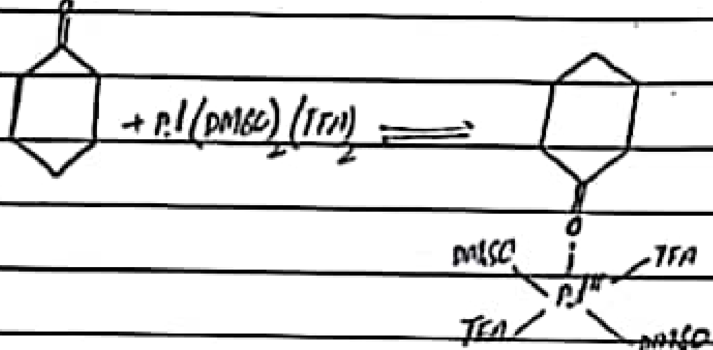
Q.NO.4

(i)(A)

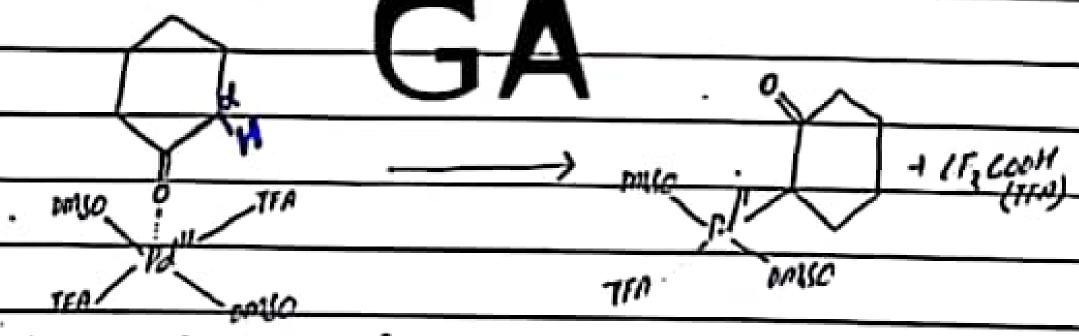


Mechanism:

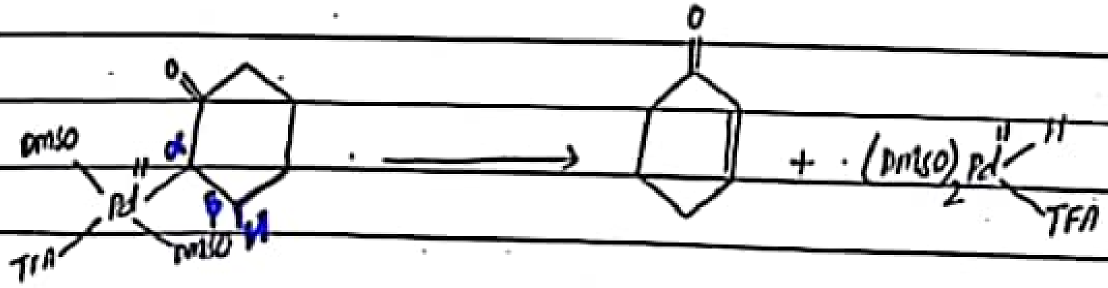
(i) Pd^{II}—cyclohexanone adduct formation



(ii) α-C-H cleavage (turnover-limiting step)



(iii) β-H-elimination



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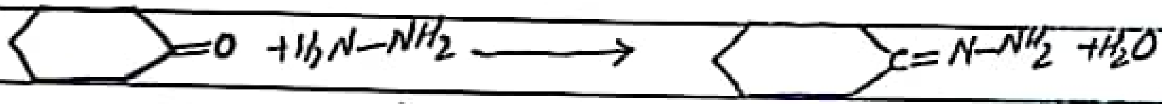
Paper : organic chemistry (2020)
Semester : 6th (CHM-317)

Q.NO.4 **GA**

Wolff Kishner Reduction ii (D)

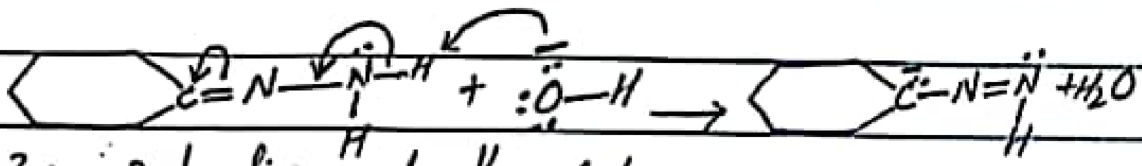
Step 1: Formation of hydrazone

The aldehyde or Ketone is subjected to hydrazine. this yields the hydrazone required for the process.



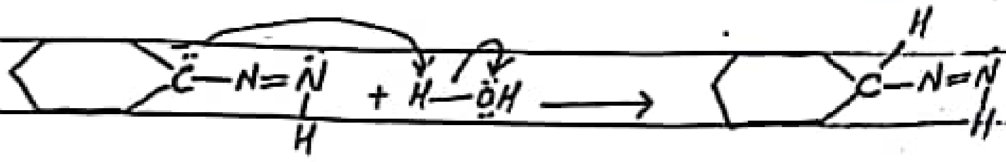
Step 2: Deprotonation of Nitrogen

The terminal nitrogen atom is deprotonated, and it proceeds to form a double bond with the neighbouring nitrogen atom. The released proton attaches itself to the hydroxide ion from the basic environment to form water.



Step 3: protonation of the Carbon

Since oxygen is more electron-withdrawing than carbon, the carbon is protonated by the water molecule as shown below.



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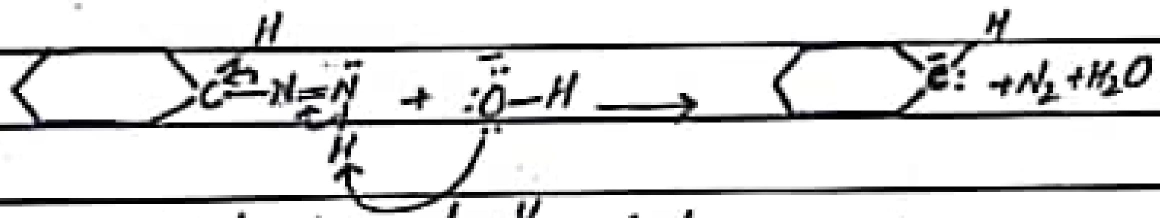
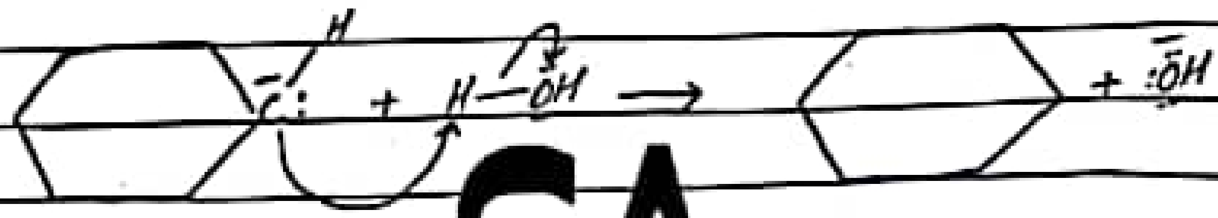
Step 4: Deprotonation of Nitrogen

The terminal nitrogen is deprotonated again, this time forming a triple bond with its neighbouring nitrogen atom. This results in the formation of a carbanion - where the two triple-bonded nitrogens are released as nitrogen gas. The ejected proton forms water along with the basic environment.



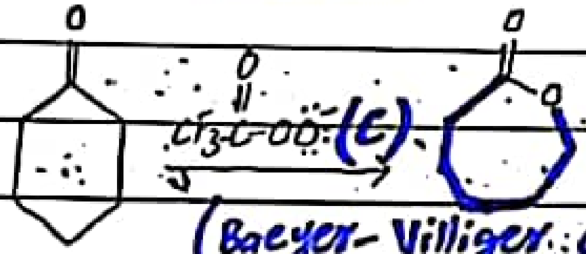
Step 5: Protonation of the Carbon

Carbanion is protonated by water, resulting in the formation of the cyclohexane hydrocarbon product.



Paper: organic chemistry (2020)
Semester: 6th (CHEM-317)

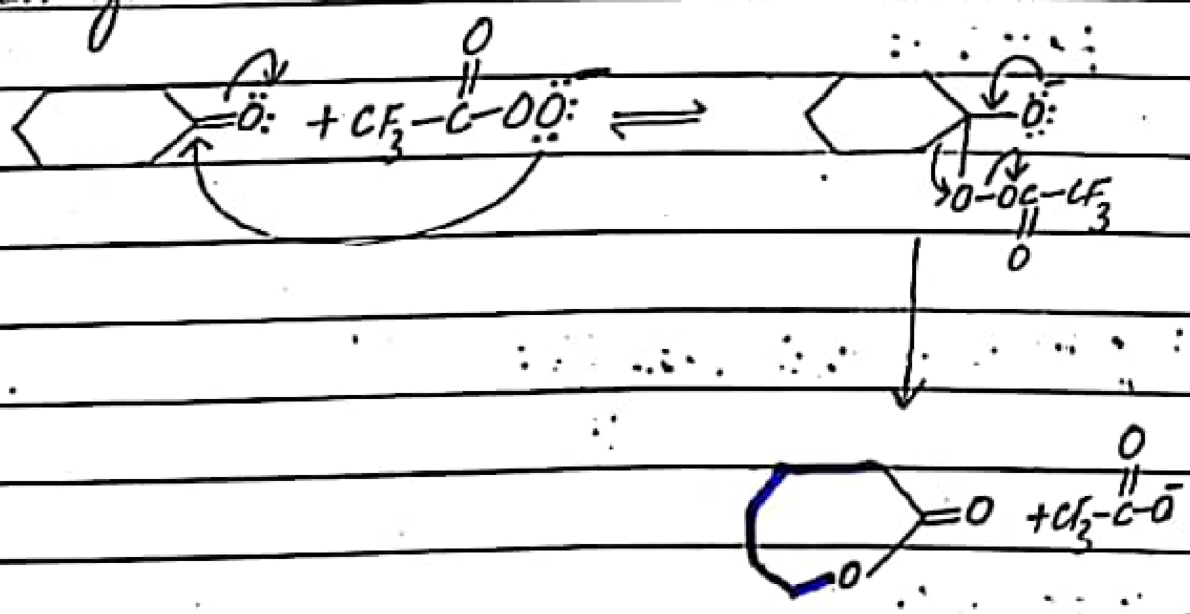
GA Q.NO.4



(Baeyer-Villiger oxidation)

Mechanism:

- The nucleophilic oxygen of the peroxyacid adds to the carbonyl carbon and forms an unstable tetrahedral intermediate with a weak O–O bond.
- As the π bond re-forms and the weak O–O bond breaks heterolytically, one of the alkyl groups migrates to an oxygen. This migration is similar to the 1,2-shifts that occur when carbocations rearrange.



Paper: organic chemistry (2020)
Semester: 6th (UCEM-317)

GA

Question No.5

(A)

Ultra-violet spectroscopy has been mainly applied for the detection of functional groups (chromophore), the extent of conjugation, detection of polynuclear compounds by comparison etc.

Some important applications of ultraviolet spectroscopy are as follows:

(a) Detection of functional groups:

The technique is applied to detect the presence or absence of the chromophore. The absence of a band at a particular wavelength may be regarded as an evidence for the absence of a particular group in the compound. A little information can be drawn from the UV spectrum if the molecule is very complicated. If the spectrum is transparent above 200nm, it shows the absence of (i) conjugation (ii) a carbonyl group (aldehydes and Ketones) (iii) benzene or aromatic compounds and also (iv) bromine or iodo.. atoms. An isolated double bond or some other atoms or groups may be present. It means that no definite conclusions can be drawn if the molecule absorbs below 200nm.

(b) Extent of conjugation:

The extent of conjugation in polyenes $R-(CH=CH)_n-R$ can be estimated. Addition in unsaturation with the increase in the number of double bonds (increase in the value of n) shifts the absorption to longer wavelength. It is found that the absorption occurs in the visible region, i.e., at about 420nm, if $n=8$ in the above polyene. Such an alkene appears coloured to the human eye.

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(c) Distinction in conjugated and non-conjugated compounds:

It also distinguishes between a conjugated and a non-conjugated compound. The following isomers can be readily distinguished since one is conjugated and the other is not.

(i) $(CH_3)_2C=CH-\overset{\overset{O}{\|}}{C}-CH_3$

(ii) $H_2C=\underset{\underset{CH_3}{|}}{C}-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$

The forbidden $n \rightarrow \pi^*$ band for the carbonyl group in the compound (i) will appear at longer wave-length compared to that for the compound (ii).

The alkyl substitution in an alkene causes a bathochromic shift. The technique is not much useful for the identification of individual alkenes.

(d) Identification of an unknown compound:

An unknown compound can be identified by comparing its spectrum with the known spectra. If the two spectra coincide, the two compounds must be identical. If the two spectra do not coincide, then the expected structure is different from the known compound.

(e) Examination of polynuclear hydrocarbons:

Benzene and polynuclear hydrocarbons have characteristic spectra in the ultra-violet and visible-region. Thus, the identification of the polynuclear hydrocarbons can be made by comparison with the spectra of known polynuclear compounds. The presence of substituents on the ring, generally, shifts the absorption maximum to longer wavelength.

(f) Elucidation of the structures of vitamins A and K:

It is useful for the elucidation of the structures of vitamin K_1 and K_2 and also those of A_1 and A_2. The ultraviolet spectra of vitamins K_1 and K_2 are due

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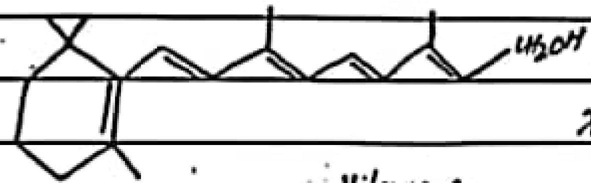
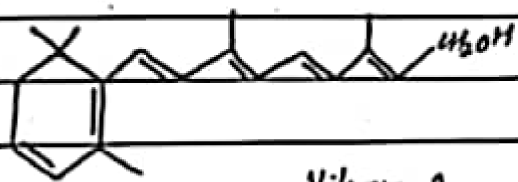
to the presence of the same chromophore, i.e., 2,3-dialkyl naphtha-quinone. The absorption maxima of this compound are 243, 249, 260, 269 and 330 nm.

The elucidation of the structures of vitamin A_1 and A_2 are possible by this technique. Vitamin A_1 absorbs at 325 nm and absorption maxima for Vitamin A_2 appear at 287 and 351 nm. The absorption maxima appear at longer wavelength for Vitamin A_2 due to the presence of additional ethylenic bond.



$\lambda_{max} = 325 nm$

Vitamin A_1

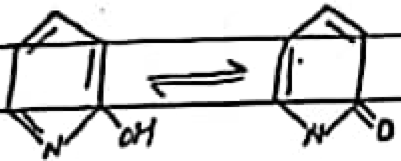


$\lambda_{max} = 325 nm$ and 351 nm

Vitamin A_2

(g) Preference over two Tautomeric forms:

If a molecule exists in two tautomeric forms, preference of one over the other can be detected by ultraviolet spectroscopy. Consider 2-hydroxy pyridine which exists in equilibrium with its tautomeric form, pyridone-2.



2-Hydroxy pyridine   Pyridone-2

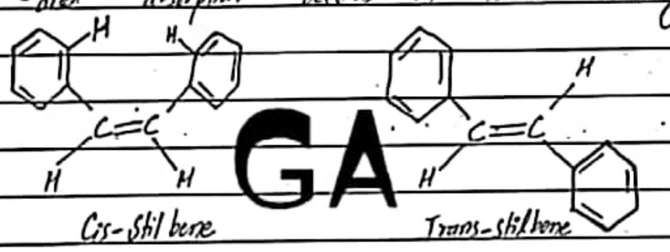
The spectra of these two compounds were found to favour pyridone-2 which is an α,β-unsaturated Ketone and clearly, the equilibrium is shifted towards the right, i.e., Pyridone-2.

(h) Identification of a compound in different solvents:

Sometimes, the structure of the compound changes with the change in the solvent. Chloral hydrate shows an absorption maximum at 290nm in hexane while the absorption disappears in the aqueous solution. Clearly, the compound contains a carbonyl group in hexane solution and its structure is $CCl_3.CHO.H_2O$ whereas in aqueous solution it is present as $CCl_3.CH(OH)_2$.

(i) Determination of configurations of Geometrical Isomers:

The results of absorption show that cis-alkenes absorb at different wavelengths as compared to their corresponding trans-isomers. The distinction becomes possible when one of the isomers is forced to be non-coplanar by steric hindrance. Thus, cis form suffer distortion and absorption occurs at lower wavelength.



Cis-Stilbene $\lambda_{max} = 283nm$

Trans-stilbene $\lambda_{max} = 295.5nm$

(j) Determination of strength of hydrogen bonding:

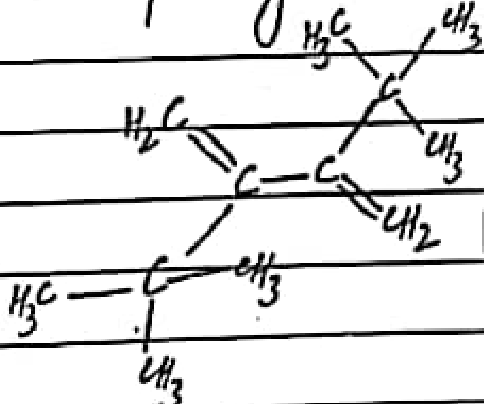
Solvents like water, C_2H_5OH etc form hydrogen bonds with the n-electrons of carbonyl oxygen. Due to this, the energy of n-electrons in the ground state is lowered depending upon the strength of hydrogen bonds. Thus, $n \rightarrow \pi^*$ transition of carbonyl compounds is shifted towards shorter wavelength. Hence, by measuring the λ_{max} of a carbonyl compound in a non-polar and polar protic solvent, the strength of hydrogen bond can be determined. Consider $n \rightarrow \pi^*$ transition of acetone in hexane (at 279nm) and that in water at 264.5nm. The blue shift by 14.5nm corresponds to an energy of 5Kcal mol^{-1}. Clearly, the strength of hydrogen bond between water and acetone is 5Kcal mol^{-1}. It is in fair agreement with the known strength of hydrogen bonds.

(K) Hindered Rotation and Conformational Analysis:

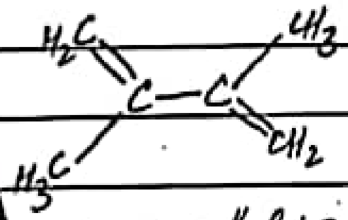
This technique is employed to study hindered rotation in sterically hindered dienes and also in ortho-substituted diphenyls.

Sterically hindered dienes:

Any deviation from coplanarity of double bonds will result in the reduction of π-orbital overlap and thus λ_{max} shifts to lower values of λ_{max} and ϵ_{max}. Consider the case of 2,3-di-tert-butyl-1,3-butadiene. In this case, there is little conjugation between two double bonds due to bulky ter-butyl groups. It shows absorption at lower wavelength (λ_{max} = 180nm) compared to 2,3-dimethyl-1,3-butadiene (λ_{max} = 225nm) in which there is no deviation from coplanarity.



2,3-Di-tert-butyl-1,3-butadiene
λ_{max} = 180nm



2,3-Dimethyl-1,3-butadiene
λ_{max} = 225nm

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